

DUSTRUNNER · DESIGN DECISIONS REGISTER

DustRunner design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 7 on 2026-10-02 (DRN-DDR-003 accepted, R10 mass restated as 16.5 kg, R14 kept and at risk, robot-side latch solenoid, ASU Photovoltaic Reliability Laboratory as first candidate pilot, appearance items, re-render); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the build plan; open decisions from DRN-DDR-001 to 003 and the review note

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, DRN-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The target modules' frame: top flange at least 21 mm wide (25 mm assumed), frame 33 mm deep, and room on top of the long-side bottom flange for the 6 mm clamp jaw	Wheel tread, hook roller, dock rail profile and frame clamp all depend on it	DRN-CAL-001 [C12]; DRN-DDR-003, P12
2	The flange bearings' bolt spacing and height (10 mm and 20 mm bores)	They set the bearing bolt holes and the clearances to the sensor brackets (4 mm) and clevises	DRN-DDR-003, P3, P5
3	The gearmotors' face hole patterns and shaft sizes	They set the housing holes and the 6 to 10 mm coupling	DRN-DDR-003, P4
4	A compression spring giving 60 N at 20 mm, about 9 mm across	Sets the hook preload	DRN-CAL-001 [C0]
5	The solenoid's pull and stroke against the pin's return spring, at 12 V for about 1 s	The robot must lift its own latch pin	DRN-DDR-003, P9
6	The charge controller's LiFePO4 profile (14.4 V) and the pack BMS's 0 °C and 45 °C charge cut-offs	Battery safety (R11)	DRN-PRC-001, Safety
7	The module maker's cleaning and frame-load guidance (60 N steady, 75 N at row entry)	R3 and R10	DRN-DDR-002, N2
8	The real masses of the bought parts	The 15.9 kg estimate and the preload rule	DRN-CAL-001 [A2], [A4]

■ Value engineering

Value-engineering target: USD 500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 573 (USD 73 over the target). Main cost drivers and savings worth trying:

- The USD 73 over the target is the parts added to make the design buildable (DRN-DDR-003): flange bearings, drive housings, couplings, roller clevises and sliders, dock brackets, clamps, ties and foot plates, the latch solenoid and beam cleats. The largest single lines are the LiFePO4 pack (USD 45), the beam (USD 30) and the dock panel (USD 22).

- Savings worth trying at quotes: cheaper flange bearings, a lighter dock, and 4 mm or pocketed truck plates (which also save about 1 kg; R10's mass limit was restated as 16.5 kg on 2026-10-02, so this is now a saving, not a fix).
- Every price is indicative; quotes are TRL 4 work and on hold (DRN-DDR-002, N3).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D12: full-width robot on 1P frames, evening cleaning gated by humidity, dock anemometer, 20 W dock charging, both wheels driven with preloaded hook rollers, 12.8 V LiFePO4 pack, 40 m reference row, R14 restated for rows of 60 m or more, budget \$500, 1P portrait 2,278 mm modules first, mechanical end stops kept, dry microfiber without airflow	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	DRN-DDR-001
2026-09-25	Start a run only below 6 m/s and abort at 8 m/s (N1); R10 allows 75 N per wheel for the row-entry transient (N2); keep <code>budget_usd</code> at \$500 as the value-engineering target and look at contingency on quotes (N3)	Amish: "i accept all your recommendations, go with them across all repos."	DRN-DDR-002
2026-09-26	DustRunner chosen for the first batch of product renders	Amish	<code>docs/REVIEW.md</code> , session 2026-09-26
2026-09-30	Write the illustrated build plan and make the design physically buildable while drawing it; keep open decisions out of the build plan, in this register	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The changes made under this instruction were accepted on 2026-10-02 (below).	DRN-DDR-003; DRN-BLD-001
2026-10-02	Design for construction accepted: the changes P1 to P16 that make the design buildable (guide roller clevises and 28 mm plate gap, sprung hook sliders, flange bearings, drive housings, beam cleats, latch solenoid, dock rail brackets, frame clamps, vertical legs and mast, three-piece end stops), as drawn	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-003, P1 to P16
2026-10-02	Robot mass: option (c), R10's mass limit restated as 16.5 kg now, with the 60 N and 75 N wheel-load limits unchanged; the robot is weighed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-003, A2; DRN-CAL-001 [A3], [A4]
2026-10-02	Payback: option (b), R14 kept at 3 years on rows of 60 m or more and marked at risk rather than restated; revisited when real quotes exist	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-003, A3; DRN-CAL-001 [I4], [I9]
2026-10-02	Latch release: option (a), the robot-side 12 V solenoid lifts its own pin, as drawn	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-003, A4
2026-10-02	Pilot site and co-design partner: a university or national lab outdoor PV test site in a dusty climate; first candidate to approach Arizona State	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-001 and 002, O1

Date	Decision	Decided by	Record
	University's Photovoltaic Reliability Laboratory in Mesa		
2026-10-02	Product appearance model deviations 1 to 6 accepted for renders, except that the dock contact bracket and bearing flanges (item 5) are taken from the constructable design, not as appearance-only parts	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , session 2026-09-26
2026-10-02	Renders: option (a), update <code>cad/src/product_model.py</code> to the constructable design and re-render the photoreal renders, card and social preview on Amish's Mac	Amish: "i approve your recommendations for all 555 open decisions."	DRN-DDR-003, Consequences